CATEGORY B

Coding Assignment

Problem 1: Priority Token Insertion

Difficulty: Easy–Medium · **Estimated time:** 20–25 minutes

TASK

The token list sometimes needs a token added at the front (priority), at the end (normal) or at a given position (rescheduled).

Implement addFirst, addLast and insertAt(int index, int token) (0-based) on your singly linked list and print the list after all operations. Write your own node class with data and next fields; do not use java.util.LinkedList.

INPUT / OUTPUT

Input: A starting list and a sequence of operations.

Output: The final list.

CONSTRAINTS

- 0 <= index <= current size
- 1 <= number of operations <= 50

EXAMPLE

```
start = 101 -> 102 -> 103
addFirst(100), addLast(104), insertAt(2, 150)

Expected Output: 100 -> 101 -> 150 -> 102 -> 103 -> 104 -> null
```

COMPLEXITY REQUIREMENTS

State the time complexity of each of the three operations.

HINT

For insertAt, walk to the node just before the index; set the new node's next first, then link the previous node to it.